WORKSHEET

Scatterplots and correlation

1 The typing speeds of a group of students, in words per minute, were measured before and after a typing course. The results were:

Original speed	32	45	29	38	37	48	51	31	26	55	41	43	36
Speed after course	38	41	38	42	32	56	62	29	38	62	50	50	47

a Draw a scatter plot to represent the data.

b Comment on the form, direction and strength of the scatter plot.

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c Calculate and interpret Pearson's correlation coefficient for these data.

2 Year 10 Maths and Science test results (in %) are recorded below.

Maths test	54	63	52	74	91	47	36	59	71	59	56	95	21	42	67	61	55	50
Science test	65	59	62	80	87	62	78	48	81	55	67	72	14	41	79	75	58	35

a Draw a scatter plot to represent the data.

b Comment on the form, direction and strength of the scatter plot.

c Calculate and interpret Pearson's correlation coefficient for these data.

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3 A group of Year 10 students was asked how much time they spent on homework each night, in minutes. Two years later, the same group was asked the same question. The results are recorded below.

Year 10	43	97	25	62	74	79	48	92	18	47	68	34	27	63	48	66	72	22
Year 12	88	125	45	98	87	129	65	104	10	76	70	47	134	109	83	95	89	31

a Draw a scatter plot to represent the data.

b Comment on the form, direction and strength of the scatter plot.

c Calculate and interpret Pearson's correlation coefficient for these data.

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4 The shoe size of each player in a soccer team is compared to the time taken for the player to sprint 100 metres (to the nearest second). The results are shown in the following table.

Shoe size	12	9	15	9	11	8	16	12	10	11	10	13	12	14
Sprint time	11	14	12	11	15	15	14	16	12	13	15	16	14	12

a Draw a scatter plot to represent the data.

b Comment on the form, direction and strength of the scatter plot.

c Calculate and interpret Pearson's correlation coefficient for these data.

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5 The heights (in cm) and weights (in kg) of a group of 17-year-old boys are as follows.

Weight	65	82	58	59	50	62	80	59	66	80
Height	173	184	177	166	163	161	173	180	179	186

a Draw a scatter plot to represent the data.

b Comment on the form, direction and strength of the scatter plot.

c Calculate and interpret Pearson's correlation coefficient for these data.

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6 The goal shooters in a netball club were asked how many goals they shot per day, on average, for practice. At the end of the season this was compared to their goal average (number of goals scored over the whole season divided by the number of games they played). The data are shown in the table below.

Average number of goals practised per day	50	100	70	50	50	100	200	150	75	80	40	120
Goal average for the season	7	17	15	12	10	15	22	25	22	15	6	19

a Draw a scatter plot to represent the data.

b Comment on the form, direction and strength of the scatter plot.

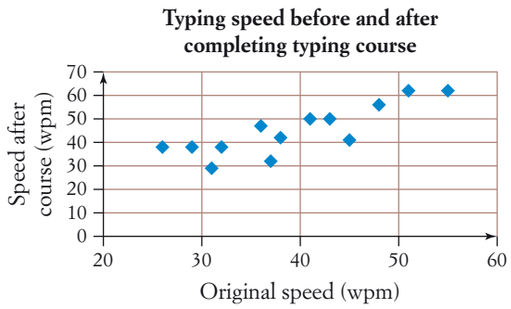
c Calculate and interpret Pearson's correlation coefficient for these data.

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Answers

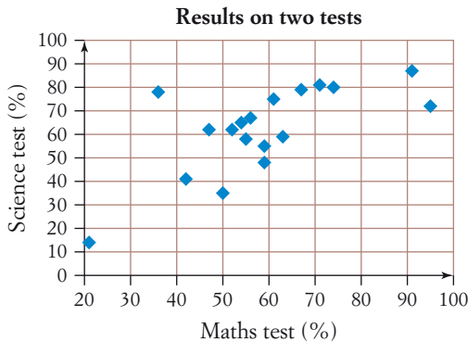
1 a



b Form: linear; direction: positive; strength: strong relationship.

c 0.84. It can be concluded that typing speed should improve after doing a typing course.

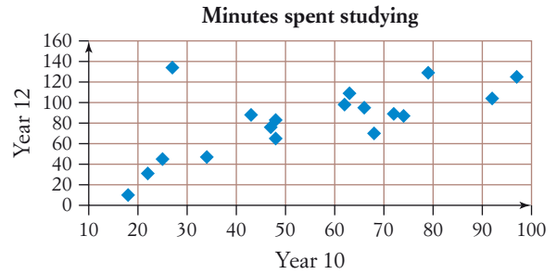
2 a



b Form: linear; direction: positive; strength: moderate relationship.

c 0.69. There is some evidence to suggest that a higher test result in Maths indicates a higher test result in Science.

3 a



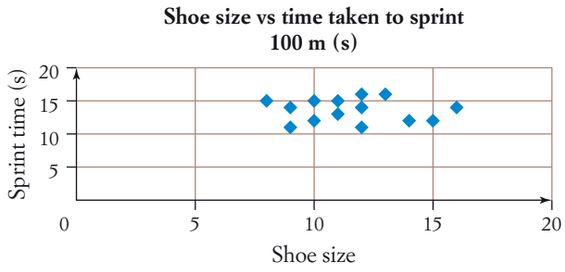
b Form: linear; direction: positive; strength: moderate relationship.

c 0.67. There is some evidence to show that an increased time spent studying in Year 10 generally indicates an increased amount of time spent studying in Year 12 or there is some evidence to show that time spent studying in Year 12 increases as time spent studying in Year 10 increases..

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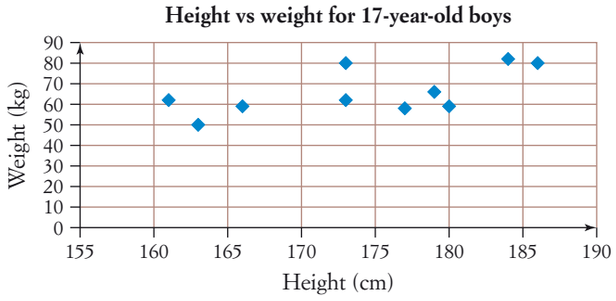
4 a



b No relationship.

c –0.07. There is no evidence to suggest that shoe size has any impact on sprinting speed for these soccer players.

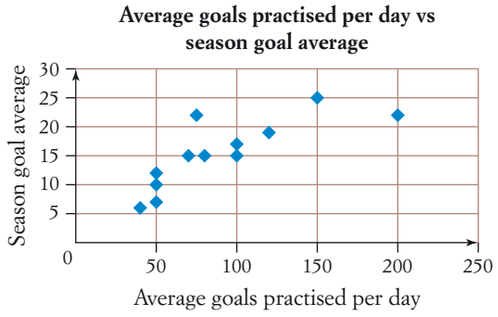
5 a



b Form: linear; direction: positive; strength: moderate relationship.

c 0.63. There is some evidence to suggest that weight increases as height increases.

6 a



b Form: linear; direction: positive; strength: strong relationship.

c 0.79. It can be concluded that a goal shooter's season goal average should increase as the average number of goals practised per day increases.

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